Supplementary Material for: *Robust Constrained Multimodal Dominance Evolution for Dynamic Weapon-Target Assignment*

This document provides supplementary material accompanying the main manuscript. It includes the complete formal mathematical model of the MO-D-POWTA problem (Section S1), the full acyclicity proof of the RCMD relation (Section S2), the TC-FCE encoding formal definition (Section S3), extended assignment visualization results (Section S4), additional experimental analyses including the per-phase neighborhood-size table and the MORI threshold sensitivity study (Section S5), and a worked example illustrating the origin of negative RMI values (Section S6). The detailed computational complexity analysis (Section S7) and a summary of the notation used throughout the paper (Section S8) are also included.

I. Complete MO-D-POWTA Mathematical Model

We present the full formal specification of the 13 hard constraints that constitute the MO-D-POWTA model. The decision variable is $x_{ijkt} \in \{0,1\}$, where $x_{ijkt} = 1$ denotes that weapon platform i fires at target j via guidance radar k during phase t. Notation follows the main manuscript (see Section II and Eqs. (1)–(3) therein).

A. Ammunition and Replenishment (Con1–Con2)

Con1 (Per-weapon ammunition capacity):

$$\sum_{j\in\mathcal{T}_t}\sum_{k\in\mathcal{R}} s_{ijkt}\, x_{ijkt} \;\le\; a_{it}, \quad \forall i \in \mathcal{W},\ t. \tag{1}$$

Here $s_{ijkt} \ge 1$ is the shot count associated with the (i,j,k) engagement. Each platform may not fire more missiles than its remaining stock a_{it}.

Con2 (Replenishment bound):

$$a_{i,t+1} \;=\; a_{it} - \sum_{j,k} s_{ijkt}\, x^{*}_{ijkt} + \delta_{it}, \quad \delta_{it} \le \Delta_i^{\max}, \tag{2}$$

where δ_{it} is the phase-t resupply quantity and $\Delta_i^{\max}$ is the platform-specific maximum resupply rate per phase.

B. Radar Channel Capacity (Con3–Con4)

Con3 (Tracking channel capacity per radar):

$$\sum_{i\in\mathcal{W}}\sum_{j\in\mathcal{T}_t} x_{ijkt} \;\le\; C_k^{\mathrm{track}}, \quad \forall k \in \mathcal{R},\ t. \tag{3}$$

Each guidance radar can simultaneously track at most C_k^{track} engagements, reflecting the finite bandwidth of phased-array fire-control systems.

Con4 (Radar assignment exclusivity):

$$\sum_{i\in\mathcal{W}} x_{ijkt} \;\le\; 1, \quad \forall j \in \mathcal{T}_t,\ k \in \mathcal{R},\ t. \tag{4}$$

A single radar channel cannot simultaneously guide two missiles toward the same target.

C. Range Feasibility and Engagement Windows (Con5–Con6)

Con5 (Weapon-target range):

$$x_{ijkt} = 0 \ \text{ if } \ d_{ij}^{(t)} > R_i^{\max}, \quad \forall i,j,k,t, \tag{5}$$

where $d_{ij}^{(t)}$ is the platform-to-target distance at phase t and $R_i^{\max}$ is the maximum engagement range of platform i.

Con6 (Engagement time window):

$$x_{ijkt} = 0 \ \text{ if } \ \tau_i^{\mathrm{react}} + d_{ij}^{(t)}/v_i \;>\; W_{jt}, \tag{6}$$

where τ_i^{react} is the platform reaction time, v_i is missile speed, and W_{jt} is the remaining engagement window for target j at phase t.

D. Engagement Multiplicity (Con7–Con8)

Con7 (Maximum missiles per target):

$$\sum_{i\in\mathcal{W}}\sum_{k\in\mathcal{R}} s_{ijkt}\, x_{ijkt} \;\le\; U_{jt}, \quad \forall j \in \mathcal{T}_t,\ t, \tag{7}$$

where U_{jt} is the maximum number of missiles assignable to target j in phase t, reflecting the diminishing marginal returns of redundant engagement.

Con8 (Minimum required engagement probability):

$$x_{ijkt} = 0 \ \text{ if } \ P_{ijk} < P_{jt}^{\min}, \quad \forall i,j,k,t, \tag{8}$$

where $P_{jt}^{\min}$ is the minimum acceptable single-shot kill probability for target j.

E. Radar Guidance Exclusivity (Con9–Con11)

Con9 (Single-target per radar channel):

$$\sum_{j\in\mathcal{T}_t} x_{ijkt} \;\le\; 1, \quad \forall i \in \mathcal{W},\ k \in \mathcal{R},\ t. \tag{9}$$

Each weapon-radar pair can guide at most one missile at a time.

Con10 (Radar detection prerequisite):

$$x_{ijkt} = 0 \ \text{ if } \ p_{kjt} < p_k^{\mathrm{th}}, \quad \forall i,j,k,t, \tag{10}$$

where p_{kjt} is the detection probability of radar k for target j at phase t and p_k^{th} is the minimum detection threshold.

Con11 (Platform-radar cooperation):

$$x_{ijkt} = 0 \;\text{ if }\; \alpha_{ik} = 0, \quad \forall i, j, k, t, \tag{11}$$

where $\alpha_{ik} \in \{0, 1\}$ is the cooperation capability matrix indicating whether platform i is compatible with guidance radar k.

F. Threat Thresholds and Platform Readiness (Con12–Con13)

Con12 (Mandatory engagement of high-threat targets):

$$\sum_{i \in \mathcal{W}} \sum_{k \in \mathcal{R}} x_{ijkt} \;\geq\; 1, \quad \forall j : Th_{jt} \geq \theta^{\text{hi}},\; t, \tag{12}$$

where θ^{hi} is the high-threat threshold (set to 0.8 in our experiments). Every high-threat target must receive at least one engagement assignment.

Con13 (Platform operational readiness):

$$x_{ijkt} = 0 \;\text{ if platform } i \text{ is not operationally ready at phase } t, \tag{13}$$

where readiness is determined by the platform's remaining ammunition ($a_{it} = 0$ renders it ineligible), maintenance status, or detected system failures.

G. Constraint Interaction Structure

The 13 constraints interact in non-trivial ways that create a structured feasibility landscape. The most significant coupling is between Con1–Con2 (ammunition) and Con9–Con11 (radar guidance exclusivity): a weapon with remaining ammunition may still be blocked from firing if all associated guidance radars are already tasked. This creates decision-space *cliques*—groups of targets that compete for the same radar channels—whose structure varies across phases as kill confirmations update target counts. TC-FCE exploits this structure by precomputing feasible combination sets $\text{Combo}(j, t)$ once per phase, reducing each genetic operator to a look-up in the feasible set rather than a constraint-satisfaction check.

II. Acyclicity of RCMD: Formal Proof

This section provides the complete proof of Theorem 1 stated in Section III-B of the main manuscript ("There is no RCMD cycle").

Proposition 1. *The* RCMD *relation is irreflexive and asymmetric.*

Proof. Irreflexivity: $d_{\text{TC}}(\mathbf{c}, \mathbf{c}) = 0 < \delta_g$, so R2 requires $\mathbf{c}$ to strictly Pareto-dominate itself—impossible. R1 requires $g(\mathbf{c}) \leq 0$ and $g(\mathbf{c}) > 0$ simultaneously—impossible. R3 requires $|g(\mathbf{c})| < |g(\mathbf{c})|$—impossible. Asymmetry: R1 is antisymmetric by definition; R2 inherits asymmetry from strict Pareto dominance; R3 is a strict order on $|g|$. □

Theorem 1. *There is no RCMD cycle* $\mathbf{c}_1 \succ_{\text{RCMD}} \mathbf{c}_2 \succ_{\text{RCMD}} \cdots \succ_{\text{RCMD}} \mathbf{c}_1$.

Proof. Assign each solution a rank $\rho(\mathbf{c}) \in \mathbb{R}$ such that $\mathbf{c}_1 \succ_{\text{RCMD}} \mathbf{c}_2$ implies $\rho(\mathbf{c}_1) < \rho(\mathbf{c}_2)$. For infeasible solutions ($g > 0$): set $\rho(\mathbf{c}) = 1 + |g(\mathbf{c})|$. R3 strictly decreases $|g|$ along each edge, so ρ is strictly decreasing—no cycle. For feasible solutions: within each modal cluster (R2), set $\rho(\mathbf{c})$ equal to the F_1 value; strict Pareto dominance strictly decreases F_1—no cycle. Across modal clusters, no edges exist—no cross-modal cycle. R1 maps infeasible ($\rho > 1$) to feasible ($\rho \leq 1$); the reverse direction requires $g(\mathbf{c}_{\text{feas}}) > 0$, which contradicts feasibility. Since ρ is strictly decreasing along every directed edge, no cycle can exist. □

III. TC-FCE Encoding: Formal Definition

Let $\text{Combo}(j, t)$ denote the set of all feasible (weapon, radar, shot-count) triplets for target j in phase t, constructed by enumerating all weapon-radar pairs within mutual detection range and multiplying by the allowed shot-count range $\{1, \ldots, L_{\max}\}$ subject to ammunition availability.

Definition 1 (TC-FCE Chromosome). *A TC-FCE chromosome for phase t is an integer vector* $\mathbf{c} = (c_1, \ldots, c_{n_t}) \in \prod_{j=1}^{n_t} \{0, \ldots, |\text{Combo}(j, t)|\}$, *where $c_j = 0$ denotes "no assignment" (target j is not engaged) and $c_j = k > 0$ denotes the k-th feasible combination in* $\text{Combo}(j, t)$.

The decoding function $\text{Dec} : \mathbf{c} \mapsto \mathbf{x}$ maps each slot to a binary assignment tensor by setting $x_{ijkt} = 1$ for the specified (weapon, radar, shot-count) triplet and zero elsewhere, then globally checking that ammunition and radar-channel sum constraints are satisfied. Any slot whose assignment would violate a global constraint is reset to $c_j = 0$ (unassigned) during decoding, maintaining feasibility without a dedicated repair step.

The Hamming distance $d_{\text{TC}}(\mathbf{c}, \mathbf{c}') = |\{j : c_j \neq c'_j\}|$ counts the number of targets whose assigned combinations differ. This metric is used both as the modal radius threshold in the RCMD definition and as the structural diversity measure (HD metric) reported in the experimental results. The TC-FCE design ensures that d_{TC} corresponds to operational differences at the individual-target level: two chromosomes at Hamming distance r disagree on the engagement plan for exactly r targets, regardless of whether those targets share weapon platforms or radar channels.

IV. Extended Assignment Visualization

Fig. 1 shows the best RCMD-EA weapon-target assignment plans across phases of instances I2 (all 3 phases) and I3 (the first 3 of 4 phases), complementing the single-phase, two-algorithm comparison in the main text with a view of how the assignment structure evolves as a single decision cycle rolls forward. The spatial visualization plots each element at its 2D geographic coordinate in the simulated theater (units: km). Weapon platforms (triangles) and guidance radars (squares) are located in the defense zone; incoming targets (circles, sized and colored by threat level) approach from the threat zone. Assignment lines connect each weapon to its assigned target, with line width proportional to the number of shots and a midpoint annotation.

V. Additional Experimental Analyses

A. Per-Metric Median Comparison Across All Instances

Fig. 2 shows the median RMI and HD achieved by all eight algorithms across instances I1–I10, with inter-quartile range

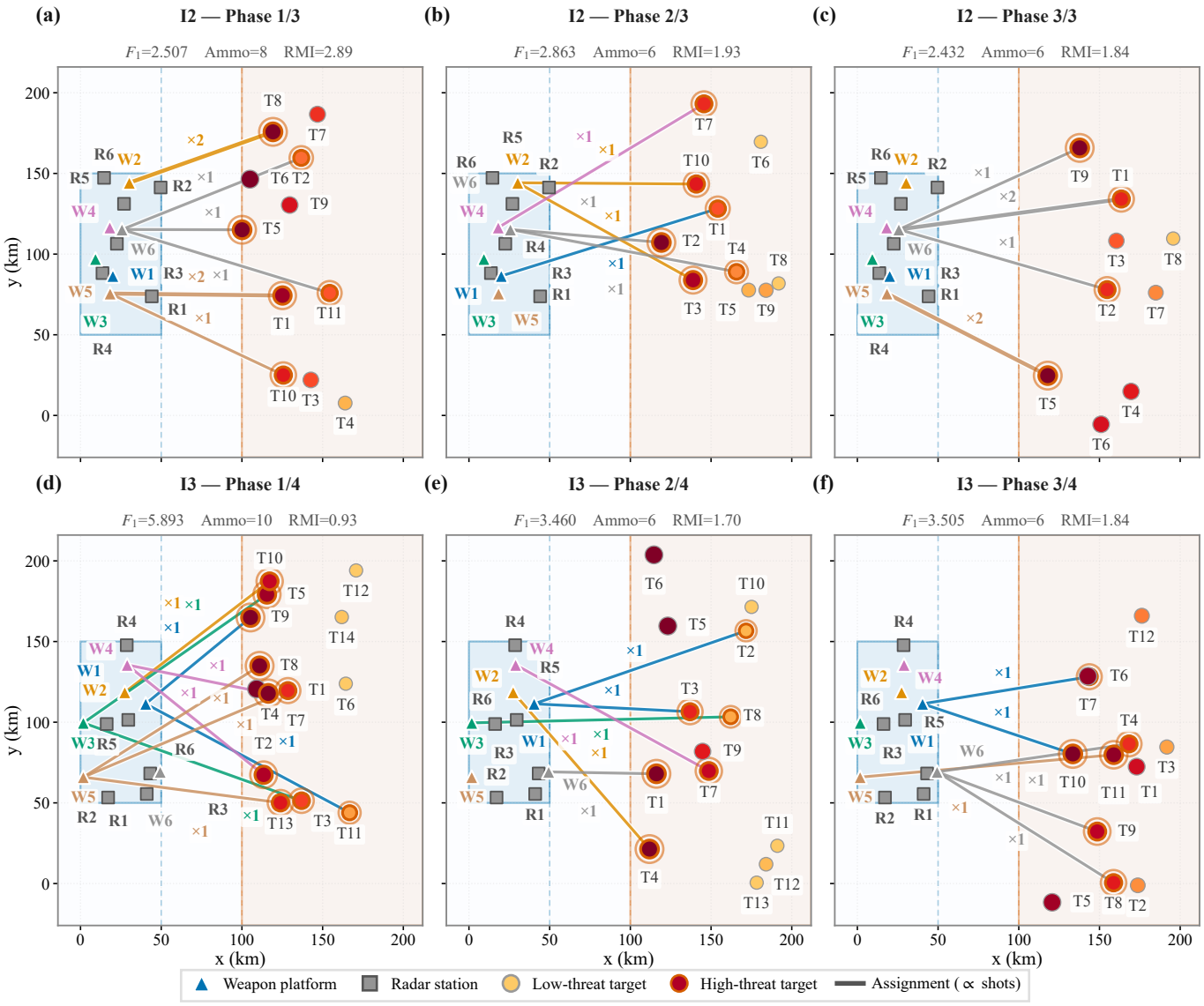


Fig. 1. RCMD-EA weapon-target assignment plans for I2 (top row, phases 1–3) and I3 (bottom row, phases 1–3 of 4). Circle size/color encodes target threat level (small/light = low, large/dark = high). Line width encodes shot count; midpoint labels show "$\times s$" for s shots.

shading over 30 independent runs (F_1 is not shown; see main text Section on Performance Metrics for why it is not used as a cross-algorithm comparison metric). RCMD-EA consistently ranks first or near-first on RMI; the multi-modal baselines (CoMMEA, HDMMODE, MMOEADC) achieve competitive HD scores but at a substantial cost to robustness.

B. RMI Advantage Heatmap

Fig. 3 summarizes the absolute difference in median RMI between RCMD-EA and each baseline, across all instances (RCMD-EA minus baseline; a percentage is not used here because RMI under the fixed-reference-point formula can be negative or near zero, which makes a percentage-improvement metric ill-defined for some baseline/instance pairs). Green cells indicate RCMD-EA wins; the two red cells (MOEA/D+TOPSIS on I6 and I10) are the sole exceptions to RCMD-EA's 68/70 win record reported in the main text.

C. RMI Distribution Across 30 Runs

Fig. 4 shows box plots of the RMI distribution (30 independent runs) for five representative algorithms on each of the ten instances, confirming that RCMD-EA's advantage is not a median artifact but holds across the full run distribution. In particular, the inter-quartile range of RCMD-EA is narrower than that of NSGA-II+TOPSIS on most instances, indicating that the robustness benefit comes with lower run-to-run variance.

D. Ablation Study: Per-Instance Detail

Table I gives the per-instance mean $\pm$ std of RMI and HD over 30 runs for the full RCMD-EA and four variants: $-$C1 (RCMD replaced by plain Pareto dominance), $-$C2 (explore_ratio set to zero, activating RCMD constraints from generation 1), $-$C3 (adaptive ϕ_t replaced by equal weighting), and

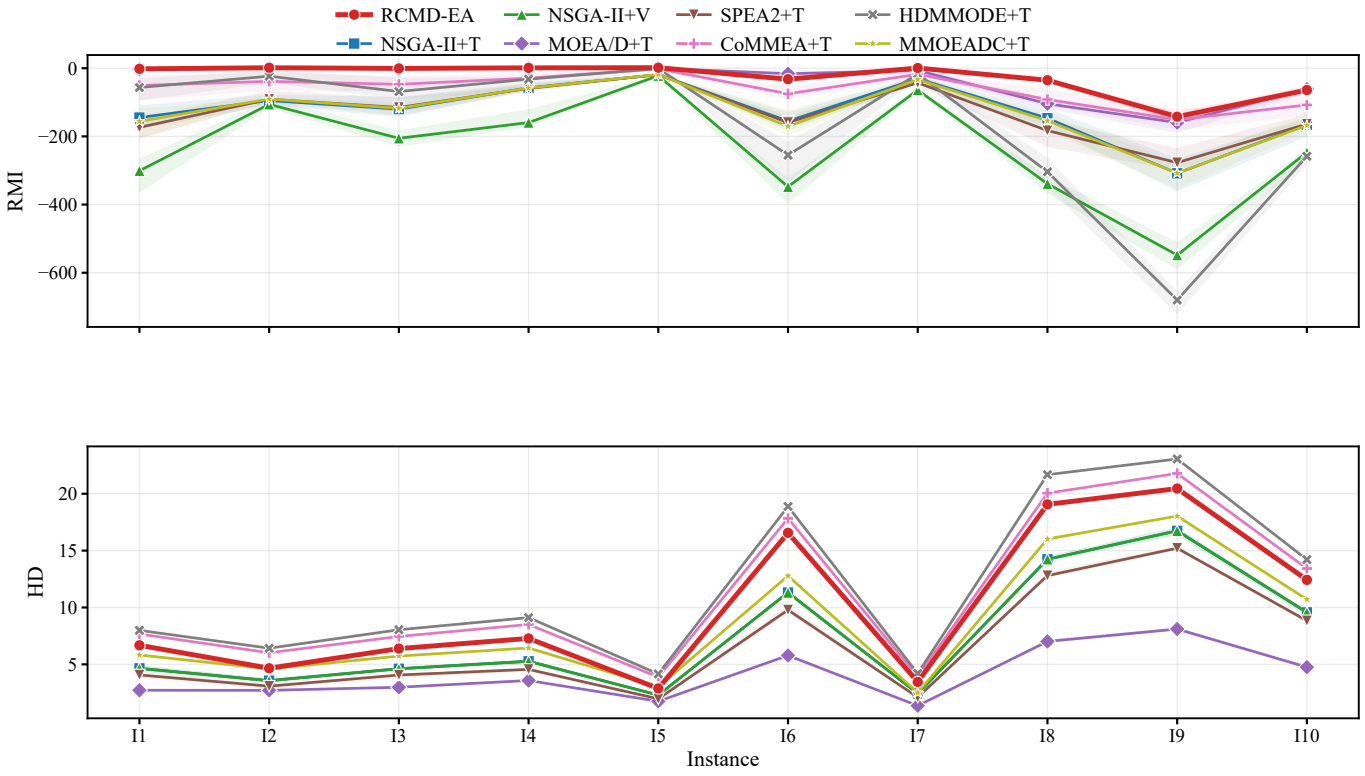


Fig. 2. Median ± IQR of RMI (top) and HD (bottom) for all 8 algorithms across I1–I10. Shaded bands span the 25th–75th percentile over 30 runs.

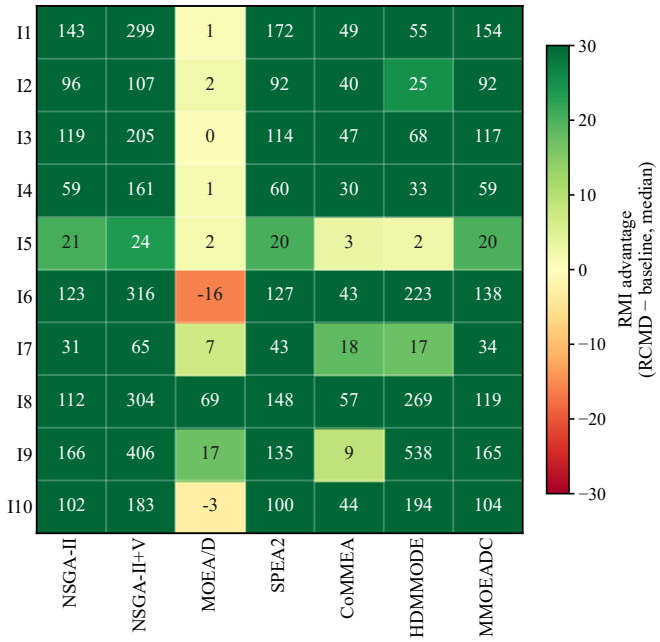


Fig. 3. RMI advantage of RCMD-EA over each baseline (absolute difference in median RMI, RCMD-EA − baseline). Green = RCMD-EA wins; red = baseline wins. RCMD-EA wins on 68 of 70 instance–baseline pairs; MOEA/D+TOPSIS on I6 and I10 are the only exceptions.

−DA (dual archive disabled). Fig. 5 visualises the same data as median ± IQR profiles, making instance-level patterns visible. The −DA variant's results are numerically indistinguishable from the full algorithm on all ten instances (lines overlap), confirming the main-text observation that the RCMD cross-modal non-domination rule already subsumes the marginal contribution of the dual archive on this benchmark.

Key observations from the per-instance breakdown: (i) −C2 degrades RMI most severely on large instances (I6, I8–I10), where the extended exploration budget is most needed to populate the Pareto front before robustness constraints are activated. (ii) −C3 shows a consistent but smaller RMI reduction across all instances, confirming that the situation-aware ϕ_t signal contributes at every scale. (iii) −DA incurs no significant RMI or HD change, validating its classification as a non-ablated design component.

E. Order-2 Hamming Neighborhood Size per Phase

Table II reports the exact neighborhood size $|\mathcal{N}_2|$ for every instance and phase, computed from the per-phase feasible-combination counts (K_j) as $|\mathcal{N}_2| = 1 + \sum_j K_j + \sum_{j_1<j_2} K_{j_1} K_{j_2}$. Three regimes are visible. First, early phases of large instances reach $|\mathcal{N}_2| > 10^6$, where RMI is estimated by adaptive Monte Carlo sampling (Eq. (3) of the main text) and remains statistically well-grounded. Second, late phases exhibit progressively smaller feasible-combination counts as range, window, and ammunition constraints eliminate engagement options, so the neighborhood shrinks toward the plan itself and RMI differences between algorithms become small by construction. Third, in the terminal phases of several instances (marked †) no target admits any feasible engagement ($K_j = 0$ for all j); the neighborhood degenerates to the plan

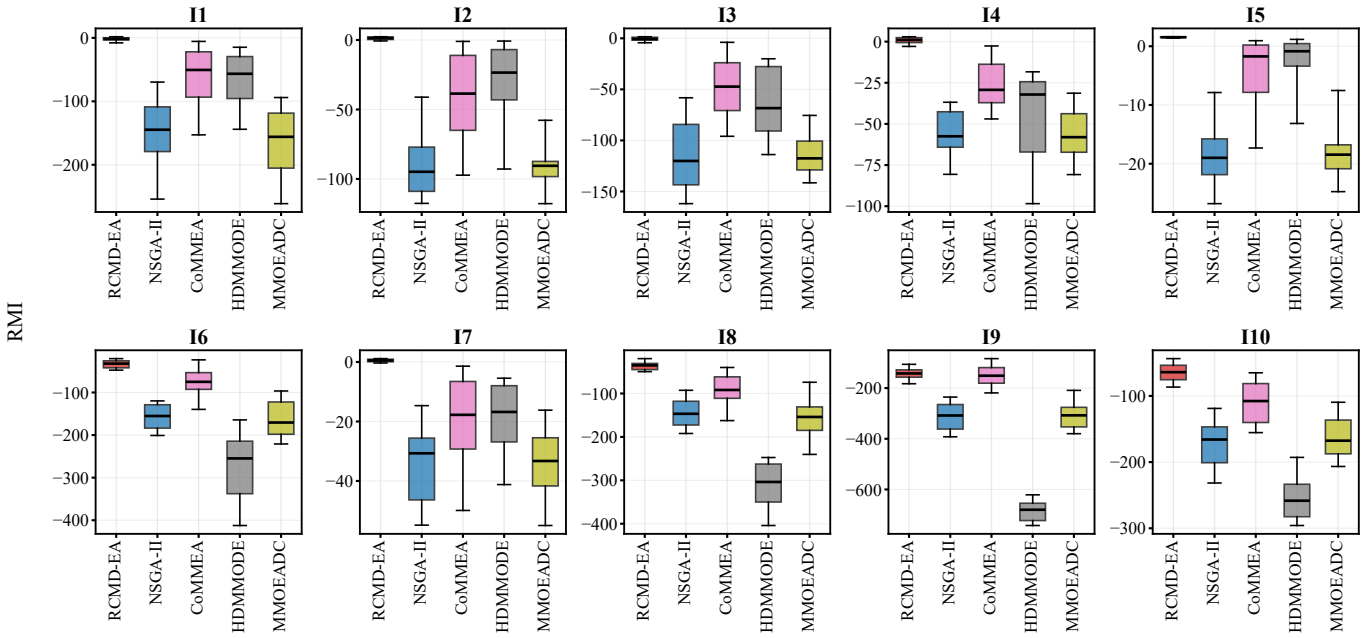


Fig. 4. RMI box plots (30 runs) for RCMD-EA, NSGA-II+TOPSIS, CoMMEA+TOPSIS, HDMMODE+TOPSIS, and MMOEADC+TOPSIS on each instance. Boxes show IQR; whiskers extend to the 5th/95th percentile. RCMD-EA medians are consistently highest or near-highest with narrower IQR on most instances.

TABLE I
ABLATION PER-INSTANCE DETAIL: MEAN ± STD OF RMI AND HD OVER 30 RUNS (REVISED V2 ABLATION SUITE). −C1 REPLACES RCMD BY PLAIN PARETO DOMINANCE; −C2 DISABLES THE EXPLORATION PHASE; −C3 FIXES $\alpha = 0.5$; −DA DISABLES THE DUAL ARCHIVE.

RMI	I1	I2	I3	I4	I5	I6	I7	I8	I9	I10
Full	−2.52±4.21	1.06±1.42	−0.85±2.09	0.59±2.09	1.52±0.08	−34.15±10.85	0.45±0.56	−36.11±10.27	−141.40±27.09	−65.52±14.56
-C1	−5.79±13.81	0.55±2.96	−1.32±2.50	−0.10±2.40	1.46±0.15	−30.15±12.22	0.24±0.89	−28.44±10.04	−137.18±27.60	−71.61±16.39
-C2	−1.86±3.01	1.78±0.88	−0.82±1.79	0.78±1.66	1.55±0.05	−66.69±17.32	0.65±0.48	−70.15±10.45	−227.83±22.05	−90.68±13.45
-C3	−17.86±14.58	−1.64±3.07	−8.26±6.14	−6.17±4.67	0.97±0.65	−48.94±16.00	−1.62±1.32	−46.77±12.33	−162.02±28.43	−76.68±14.84
-DA	−5.10±7.71	0.53±2.29	−1.49±2.96	0.63±1.76	1.50±0.12	−33.92±11.08	0.61±0.42	−30.26±8.86	−127.20±21.56	−63.40±12.69
HD	**I1**	**I2**	**I3**	**I4**	**I5**	**I6**	**I7**	**I8**	**I9**	**I10**
Full	6.69±0.28	4.67±0.34	6.39±0.18	7.29±0.13	2.90±0.10	16.56±0.17	3.45±0.08	19.08±0.20	20.49±0.13	12.43±0.09
-C1	6.15±0.21	4.77±0.17	6.10±0.10	7.08±0.09	3.10±0.07	14.69±0.26	3.27±0.05	17.19±0.27	18.82±0.19	11.66±0.14
-C2	8.01±0.09	6.07±0.16	7.72±0.09	8.91±0.08	3.95±0.07	18.47±0.07	4.03±0.08	20.90±0.05	22.45±0.10	13.77±0.05
-C3	6.69±0.28	4.67±0.34	6.39±0.18	7.29±0.13	2.90±0.10	16.56±0.17	3.45±0.08	19.08±0.20	20.49±0.13	12.43±0.09
-DA	5.20±0.29	4.04±0.25	5.38±0.27	6.43±0.28	2.61±0.12	15.45±0.28	3.08±0.08	18.47±0.30	19.80±0.23	12.18±0.15

itself, no intercept is possible, and $\mathrm{RMI} \equiv F_1^{\mathrm{ref}} - F_1(\mathbf{c}) = 0$ for every algorithm. This table complements the phase-by-phase analysis of Section 5.2 in the main text and allows readers to identify exactly where RMI remains a discriminative metric.

F. Sensitivity of the MORI Feasibility Threshold

The main text defines robustness feasibility relative to the population mean of MORI (Eq. (7) of the main text). Because the threshold is population-derived, a natural concern is whether the specific choice of estimator matters, particularly for smaller populations where the mean is a noisy statistic. We therefore re-ran the full algorithm with three alternative threshold definitions: the 40th and 60th population percentiles (MORI-p40, MORI-p60, relaxing and tightening the feasibility pressure, respectively) and an absolute threshold $\mathrm{MORI} \geq 0$ (MORI-abs0, requiring a non-negative average range-normalized neighborhood improvement).

Table III reports the results over 30 runs per instance. The two percentile variants are statistically indistinguishable from the default mean threshold (MORI-p40: no significant difference on any instance; MORI-p60: significant only on I1 and I2), indicating that the mechanism is robust to the exact location of a population-derived threshold. The absolute threshold, in contrast, significantly degrades RMI on the four largest instances (I6, I8–I10; mean −42.71 vs −27.69) and reduces mean HD from 9.99 to 8.33: an absolute cut-off ignores the scale of the current population's robustness distribution and applies an inappropriately loose filter early (when most solutions are fragile) and an inappropriately strict one late, destabilizing the selection pressure that the adaptive design intends.

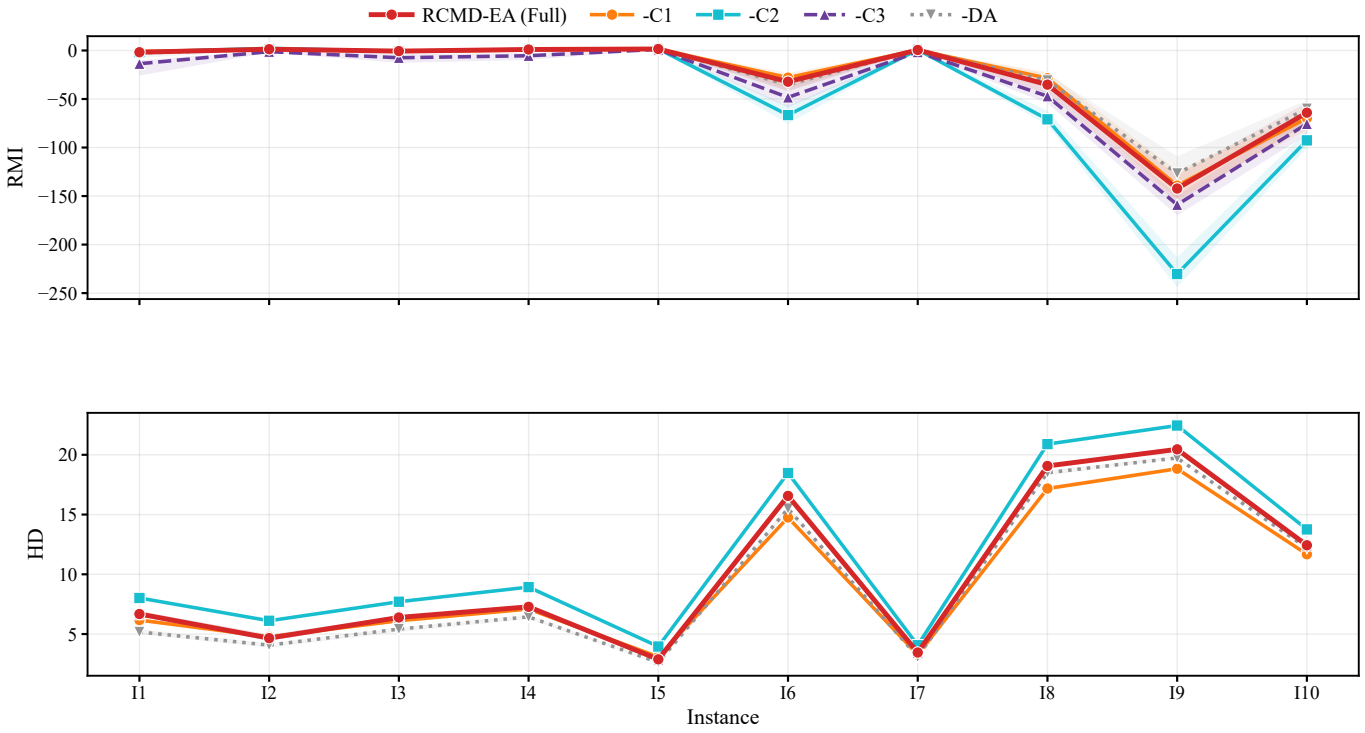


Fig. 5. Per-instance median ± IQR of RMI (top) and HD (bottom) for RCMD-EA (Full), −C1, −C2, −C3, and −DA variants over 30 runs. Shaded bands span the 25th–75th percentile. The −DA line (grey dotted) overlaps the Full line throughout, confirming that the dual archive contributes negligibly on this benchmark.

TABLE II

EXACT ORDER-2 HAMMING NEIGHBORHOOD SIZE $|\mathcal{N}_2|$, COMPUTED FROM THE PER-PHASE FEASIBLE-COMBINATION COUNTS AS $|\mathcal{N}_2| = 1 + \sum_j K_j + \sum_{j_1<j_2} K_{j_1} K_{j_2}$ (SECTION S7 OF THIS DOCUMENT). †DEGENERATE PHASES IN WHICH NO TARGET ADMITS ANY FEASIBLE ENGAGEMENT ($K_j = 0\ \forall j$): THE NEIGHBORHOOD REDUCES TO THE PLAN ITSELF AND RMI ≡ 0 REGARDLESS OF THE ALGORITHM. THESE PHASES EXPLAIN THE LATE-PHASE RMI CONVERGENCE TOWARD ZERO DISCUSSED IN SECTION 5.2 OF THE MAIN TEXT.

Instance	T	$\|\mathcal{N}_2\|$ per phase (k=10^3, M=10^6)									
I1	2	163k	17k								
I2	3	54k	109k	15k							
I3	4	192k	41k	52k	2k						
I4	5	294k	143k	62k	6k	14k					
I5	8	282k	103k	50k	19k	18k	1†	1†	1†		
I6	5	1.8M	1.1M	377k	122k	42k					
I7	10	937k	397k	84k	32k	261	1k	5	1†	1†	1†
I8	6	9.2M	3.6M	2.2M	528k	234k	88k				
I9	8	17.9M	8.4M	3.5M	1.6M	387k	59k	14k	5		
I10	10	11.8M	4.2M	3.0M	831k	196k	22k	294	3k	23k	1†

VI. WORKED EXAMPLE: NEGATIVE RMI FROM CONSTRAINT-VIOLATING NEIGHBORS

This section provides the toy example referenced in Section 2.5 of the main text, illustrating (i) how the constraint-violation penalty enters the neighbor evaluation and (ii) why RMI can be strongly negative for plans that sit at the constraint boundary.

Setup. Consider a single-phase instance with two targets, one weapon platform carrying three missiles, and one guidance radar. Both targets have $\eta_{jt} Th_{jt} = 1$, the kill probability is $P = 0.6$ for every shot, and each target may receive at most $U = 2$ shots (Con7). The TC-FCE encoding therefore gives each target slot three values: 0 (no intercept), 1 (one shot), 2 (two shots), i.e., $K_1 = K_2 = 2$. The zero-interception reference is $F_1^{\text{ref}} = \eta_1 Th_1 + \eta_2 Th_2 = 2$. For readability we use neighborhood order $r = 1$; the mechanism is identical for the $r = 2$ used throughout the experiments.

All plans and their penalized F_1. Table IV enumerates all nine plans. Every plan is individually feasible w.r.t. the per-target combination sets, but the aggregate plan $(2, 2)$ fires four missiles against a magazine of three, violating Con1 by one unit and incurring a penalty of $\lambda = 10^3$ (Section 2.5 of the main text), giving $F_1 = 1000.32$.

RMI of a boundary plan versus a slack plan. Plan A $= (2, 1)$ consumes the entire magazine. Its order-1 neighbors are $\{(0, 1), (1, 1), (2, 0), (2, 2)\}$, which include the Con1-violating plan $(2, 2)$:

$$\text{RMI(A)} = 2 - \tfrac{1.40+0.80+1.16+1000.32}{4} = -248.92.$$

TABLE III
SENSITIVITY OF THE MORI FEASIBILITY THRESHOLD: PER-METRIC MEANS ACROSS I1–I10 OVER 30 RUNS PER INSTANCE (RMI/HD AVERAGED OVER PHASES; HV = MEAN PER-PHASE HYPERVOLUME). SIGNIFICANCE VS. THE DEFAULT MEAN THRESHOLD (WILCOXON SIGNED-RANK ON RMI): † $p < 0.05$ ON 4/10 INSTANCES; NO MARK = NO SIGNIFICANT DIFFERENCE ON ANY INSTANCE.

Threshold	RMI↑	HD↑	HV↑
Mean (default)	$\mathbf{-27.69}$	**9.99**	184.84
40th percentile	-27.71	9.93	**185.79**
60th percentile	-27.92	9.75	185.69
Absolute (≥ 0)	-42.71†	8.33	183.42

TABLE IV
WORKED EXAMPLE: PENALIZED F_1 OF ALL NINE PLANS OF A TOY INSTANCE (TWO TARGETS, ONE PLATFORM WITH THREE MISSILES, $U = 2$, $P = 0.6$, $\eta_{jt} Th_{jt} = 1$; $F_1^{\text{ref}} = 2$). •: ORDER-1 NEIGHBORS OF PLAN A$= (2, 1)$; ○: ORDER-1 NEIGHBORS OF B$= (1, 0)$; (A)/(B) MARK THE PLANS THEMSELVES. SLOT VALUES: 0 = NO INTERCEPT, 1 = ONE SHOT, 2 = TWO SHOTS.

Plan (c_1, c_2)	Shots	Con1 overshoot	F_1 (penalized)	$\in \mathcal{N}_1(\text{A})$	$\in \mathcal{N}_1(\text{B})$
(0,0)	0	–	2.0000		○
(0,1)	1	–	1.4000	•	
(0,2)	2	–	1.1600		
(1,0)	1	–	1.4000		(B)
(1,1)	2	–	0.8000	•	○
(1,2)	3	–	0.5600		○
(2,0)	2	–	1.1600	•	○
(2,1)	3	–	0.5600	(A)	
(2,2)	4	1	1000.3200	•	

Plan B $= (1, 0)$ leaves two missiles of headroom. Its neighbors $\{(0, 0), (2, 0), (1, 1), (1, 2)\}$ are all feasible:

$$\text{RMI(B)} = 2 - \tfrac{2.00+1.16+0.80+0.56}{4} = +0.87.$$

Both plans have identical nominal effectiveness along the neighbor set on the feasible members, yet their robustness differs by ≈ 250 RMI units purely because one perturbation step away from plan A crosses the ammunition boundary. This is precisely the fragility signal that the RCMD feasibility layer is designed to penalize during search, and it explains the large negative values observed for constraint-boundary-seeking baselines in Table 3 of the main text (e.g., -688.7 for HDMMODE+TOPSIS on I9, which corresponds to an average overshoot of ≈ 0.69 units per neighbor at the sampled neighborhood).

VII. COMPUTATIONAL COMPLEXITY: FULL DERIVATION

This section expands the complexity summary given in Section 3.6 of the main text.

Neighborhood size. The order-r Hamming neighborhood of a TC-FCE chromosome depends not only on the number of target slots n_t but also on the number of feasible engagement combinations per slot. Let $K_j = |\text{Combo}(j, t)|$ denote the size of the pre-computed feasible-combination set of target j (Section II of the main text). Changing exactly k slots yields $\sum_{|S|=k} \prod_{j \in S} K_j$ alternatives, so

$$\begin{aligned} |\mathcal{N}_2(\mathbf{c})| &= 1 + \sum\nolimits_j K_j + \sum\nolimits_{j_1 < j_2} K_{j_1} K_{j_2} \\ &\leq 1 + n_t \bar{K} + \tbinom{n_t}{2} \bar{K}^2, \end{aligned} \tag{14}$$

where $\bar{K}$ is the mean combination count and the bound is attained when all slots share the same $K_j = \bar{K}$. Across I1–I10, $\bar{K}$ ranges from ≈ 0 (late phases with no feasible engagement left) to ≈ 93, and the resulting $|\mathcal{N}_2|$ spans from a single member (degenerate late phases) to $\sim 1.8 \times 10^7$ (I9, phase 1); the exact per-phase values are reported in Section S5. The dominant cost per generation is $\mathcal{O}(N^2 d + N \cdot S^* \cdot n_t \cdot m \cdot L)$, where d is chromosome length, N is population size, and the second term is the RMI evaluation of N solutions, each requiring S^* neighbor decodings ($\mathcal{O}(n_t)$) and batched penalized F_1/F_2 evaluations ($\mathcal{O}(n_t \cdot m \cdot L)$ per neighbor). The sampling budget S^*, not the raw neighborhood size, controls the per-solution RMI evaluation cost. The adaptive S^* (Eq. (3) of the main text) takes values between the accuracy floor $S^* = 67$ and the cap $S^* = 500$, the latter binding in all early phases of the large instances I8–I10. This prevents quadratic blowup for large instances while maintaining a 90% confidence accuracy guarantee.

The TC-FCE decoding step is $\mathcal{O}(n_t \cdot m \cdot L)$ per solution per generation, which is linear in all problem dimensions and contributes negligibly to the total runtime. The non-dominated sorting under RCMD dominance uses the standard fast non-dominated sort of NSGA-II with one additional robustness-feasibility check per comparison, adding only an $\mathcal{O}(N^2)$ term that is dominated by the RMI evaluation.

In practice, the dominant cost across all tested instances is the RMI evaluation. For large instances such as I9 (67 active targets in phase 1, $|\mathcal{N}_2| \approx 1.8 \times 10^7$), the sampling cap limits each RMI evaluation to at most 500 perturbation samples, resulting in a per-generation cost that is linear in N at that cap.

The dual-archive update requires maintaining two archive sets of size $\lfloor N/2 \rfloor$ and $\lceil N/2 \rceil$ respectively, with Hamming-maximin computation for $\mathcal{D}_A$. Hamming distance between two chromosomes is $\mathcal{O}(d)$, so the full archive update is $\mathcal{O}(N \cdot d)$ per generation, contributing less than 5% of total runtime in experiments.

VIII. NOTATION SUMMARY

Table V summarizes the symbols used throughout the paper, in order of appearance; each symbol is also defined at its first occurrence in the main text.

TABLE V
NOTATION USED IN THIS PAPER.

Symbol	Meaning
m, L	weapon platforms; guidance radars
T	number of phases
n	total distinct targets over the engagement
n_t	active targets in phase t
$\mathcal{T}_t$, $\mathcal{W}$, $\mathcal{R}$	active target / weapon / radar index sets
Th_{jt}, η_{jt}	threat level; MRFDQI tracking quality of target j
P_{ijk}	kill probability of platform i on target j via radar k
a_{it}	ammunition available to platform i in phase t
C_k^{track}	tracking-channel capacity of radar k
$\mathbf{c} = (c_1, \ldots, c_{n_t})$	TC-FCE chromosome (one slot per target)
K_j	feasible-combination count of slot j ($\lvert\mathrm{Combo}(j,t)\rvert$)
x_{ijkt}	decoded assignment tensor
$d_{\mathrm{TC}}(\cdot,\cdot)$	TC-Hamming distance (differing slots)
$\mathcal{N}_r(\mathbf{c})$	order-r Hamming neighborhood
F_1, F_2	residual threat; ammunition consumption
F_1^{ref}, F_2^{ref}	zero-interception / full-allocation references
RMI_q	fixed-reference robustness of objective q (main text Def. 2)
RMI	reported RMI ($= \mathrm{RMI}_1$ of the post-selected plan)
$\widetilde{R}_q$	population range of F_q
MORI, $\overline{\text{MORI}}$	modal robustness index; its pop. mean (main text Eq. (7))
$g(\mathbf{c})$	robustness infeasibility ($\overline{\text{MORI}} - \text{MORI}(\mathbf{c})$)
w_g, ϕ_t	intra-generation search signal; situational signal
δ_g	modal-cluster radius
$\succ_{\text{RCMD}}$, R1–R3	RCMD dominance and its three rules
$\mathcal{Q}_A$, $\mathcal{D}_A$	quality archive; diversity archive
N, $G_{\max}$	population size; generations per phase
S^*	adaptive RMI sample size (main text Eq. (3))
ε_{rel}, δ_{prob}	RMI accuracy / confidence parameters
p_c, p_{mut}, p_{guided}	crossover / mutation / guided rates
ε	tie-band threshold of post-selection
HD, HV	mean pairwise Hamming diversity; hypervolume